

Part 3: Critical Thinking

Ethics & Bias

Question 1: How might biased training data affect patient outcomes in the case study?

Impact of Biased Training Data on Hospital Readmission Predictions:

1. Underrepresentation of Minority Groups

- If training data lacks diverse demographics, the model performs poorly for underrepresented populations
- **Patient Impact:** Minority patients incorrectly classified as low-risk miss critical post-discharge interventions
- **Result:** Higher readmission rates for these groups, perpetuating health disparities

2. Socioeconomic Bias

- Data from well-resourced hospitals doesn't reflect underserved communities
- Model underestimates risk for patients with limited access to care, transportation, or medications
- **Result:** Highest-need patients receive least support, leading to preventable readmissions

3. Historical Healthcare Disparities

- Past discriminatory practices embedded in historical data
- Model learns that certain groups "need less intervention"
- **Example:** If Black patients historically received fewer home health referrals (due to bias), model recommends fewer interventions even when clinically needed

4. Measurement Bias

- Medical devices have different accuracy across populations (e.g., pulse oximeters less accurate on darker skin)
- Inaccurate measurements lead to incorrect risk assessments
- **Result:** Missed warning signs, delayed treatment, worse outcomes

5. Proxy Discrimination

- Using zip code, insurance type as proxies for protected characteristics
- Patients from certain areas automatically flagged as lower risk
- **Result:** Systematic under-serving of vulnerable populations

Real-World Example:

Optum's algorithm used healthcare costs to predict health needs. Black patients had lower costs (less access) but higher actual needs. The algorithm systematically underestimated their risk, denying them care management programs.

Question 2: Suggest 1 strategy to mitigate this bias

Strategy: Fairness-Aware Model Development with Stratified Validation

Implementation:

1. Diverse Data Collection

- Actively collect data from diverse patient populations (race, ethnicity, socioeconomic status, geography)
- Partner with community health centers serving underrepresented groups
- Use stratified sampling to ensure balanced representation

2. Subgroup Performance Evaluation

- Evaluate model separately for each demographic group
- Track metrics: Accuracy, Recall, Precision, F1 Score per group
- Set minimum thresholds: F1 Score ≥ 0.75 for every subgroup
- Flag groups performing >10% worse than best-performing group

Example Evaluation:

Group	Accuracy	Recall	Status
White	0.82	0.78	✓ Pass
Black	0.79	0.74	Review
Hispanic	0.80	0.76	✓ Pass
Low Income	0.77	0.72	✗ Fail - Needs Improvement

3. Apply Fairness Techniques

- **Reweighting:** Assign higher weights to underrepresented groups during training
- **Calibration:** Adjust prediction thresholds per group to equalize outcomes
- **Feature Auditing:** Remove biased proxies (zip code → area deprivation index)

4. Continuous Monitoring

- Track performance monthly by demographic group
- Conduct quarterly bias audits
- Retrain model with new, diverse data
- Establish bias response team to address issues quickly

Why This Works:

- Addresses bias at data, model, and deployment stages
- Measurable with clear thresholds
- Sustainable through continuous monitoring
- Ensures equitable care across all patient populations

Trade-offs

Question 1: Discuss the trade-off between model interpretability and accuracy in healthcare

The Fundamental Trade-off:

High Interpretability → Moderate Interpretability → Low Interpretability

Low Accuracy → Moderate Accuracy → High Accuracy

Logistic Regression (78% accuracy)	→	Random Forest (82% accuracy)	→	Deep Neural Networks (86% accuracy)
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Arguments for Interpretability:

1. **Clinical Trust:** Physicians need to understand WHY a model makes recommendations
 - "Previous admissions > 2 AND Diabetes = Yes → High Risk" is actionable
 - Black-box predictions create hesitation and low adoption
2. **Legal Requirements:** Healthcare decisions must be explainable for liability and informed consent
3. **Error Detection:** Easy to identify when model uses spurious correlations (e.g., relying on hospital ID instead of clinical features)
4. **Patient Communication:** Patients have right to understand their risk factors
 - "Your risk is high due to heart failure and living alone" enables better care planning

Arguments for Accuracy:

1. **Lives Saved:** Even 2% accuracy improvement saves lives at scale

- Hospital with 10,000 discharges/year: 80% vs 82% = 200 more patients correctly identified

2. **Complex Patterns:** Many conditions require modeling intricate interactions

- Sepsis prediction needs temporal patterns in dozens of variables
- Simple models miss critical non-linear relationships

3. **Personalized Medicine:** High-dimensional problems (genomics, imaging) require complex models

The Balanced Approach:

Hybrid Strategy - Best of Both Worlds:

1. Use Random Forest for predictions (80-82% accuracy)
2. Add SHAP explanations showing top 5 risk factors per patient
3. Require clinical review for all high-risk predictions

Example:

- Random Forest predicts: High Risk (85% confidence)
- SHAP explains: "Top factors: 3 previous admissions, diabetes, no caregiver, age 75, long hospital stay"
- Clinician validates and creates personalized discharge plan

Result: High accuracy WITH sufficient interpretability for clinical adoption

Recommendation for Readmission Prediction:

- **Primary Model:** Random Forest (good balance)
 - **Explanation:** SHAP values for each prediction
 - **Human Oversight:** Case manager reviews all high-risk cases
 - **Achieves:** >80% accuracy + interpretability + clinical trust
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Question 2: If the hospital has limited computational resources, how might this impact model choice?

Resource Constraints:

- No GPU (CPU only)
- Limited RAM (8-16 GB)
- Need real-time predictions (< 2 seconds)
- Small IT staff
- No cloud budget

Impact on Model Selection:

Models to AVOID:

- ❌ Deep Neural Networks (requires GPU, slow on CPU, GBs of RAM)
- ❌ Large Ensembles (multiplies resource needs)
- ❌ Complex models with 100s of trees

Models RECOMMENDED:

1. Logistic Regression

- Training: Seconds
- Inference: < 1ms
- Memory: < 1 MB
- Accuracy: 78%
- ✓ Perfect for real-time predictions

2. Random Forest (Optimized)

- Training: 2-3 minutes
- Inference: 5-10ms
- Memory: 100-500 MB
- Accuracy: 80-82%
- ✓ Best balance for limited resources

```
RandomForestClassifier(  
    n_estimators=50,          # Reduce from 100+  
    max_depth=10,            # Limit depth  
    min_samples_leaf=10,  
    n_jobs=-1                # Use all CPU cores  
)
```

3. LightGBM (Optimized)

- Training: 2-3 minutes
- Inference: 10-20ms
- Memory: 200-800 MB
- Accuracy: 81-83%
- ✓ CPU-optimized, excellent efficiency

Optimization Techniques:

1. Feature Selection

- Use only top 15 most important features (instead of 50)
- Faster inference, lower memory
- Accuracy loss: < 2%

2. Batch Processing

- Run predictions nightly for all current patients
- Store results in database
- At discharge: Instant lookup (< 100ms)
- Real-time prediction only for new admissions

3. Model Compression

- Quantization: Convert 32-bit to 8-bit (4x smaller, 2x faster)
- Minimal accuracy loss (< 1%)

Real-World Example: Rural Hospital

Constraints:

- Single server, no GPU, 16 GB RAM
- 2,000 discharges/year
- Need: < 2 second response time
- Budget: \$0 for new hardware

Solution:

1. Selected top 15 features (from 50)
2. Used Random Forest (50 trees, depth=5)
3. Batch predictions nightly
4. Database lookup at discharge

Results:

- Accuracy: 80% (exceeds 78% target)
- Response time: < 1 second
- Cost: \$0 (existing hardware)
- Maintenance: 2 hours/week

Trade-off Accepted: 3-5% lower accuracy than deep learning, but actually deployable and sustainable

Key Principle:

"A good model that runs is better than a perfect model that doesn't."

Recommendations:

1. Start with LightGBM or Random Forest (optimized)

- 2. Focus on data quality over model complexity
- 3. Implement efficient deployment (batch + caching)
- 4. Use free/open-source tools (Python, scikit-learn)

Cost-Benefit Comparison:

Model	Inference Time	Memory	Accuracy	Hardware Cost	Feasible?
Logistic Regression	< 1 ms	1 MB	78%	\$0	✓ Yes
Random Forest	8 ms	250 MB	80%	\$0	✓ Yes
LightGBM	12 ms	300 MB	81%	\$0	✓ Yes
Neural Network	100 ms	2 GB	84%	\$5,000 GPU	✗ No

Bottom Line:

Limited resources should not prevent AI implementation. Smart model selection and optimization enable resource-constrained hospitals to achieve clinically meaningful results (78-82% accuracy) that improve patient outcomes.

Conclusion

Successful healthcare AI requires:

- **Ethical vigilance** to ensure fairness across all patient groups
- **Strategic trade-offs** balancing interpretability, accuracy, and resources
- **Practical solutions** that work within real-world constraints
- **Patient-centered approach** prioritizing outcomes over technical perfection